CUSTOMER DATA ANALYSIS

Tools Used: MYSQL AND MICROSOFT POWER BI

Project Focus: Customer Shopping Behaviour, Gender, Age Groups, Product Categories, Revenue and Payment Methods

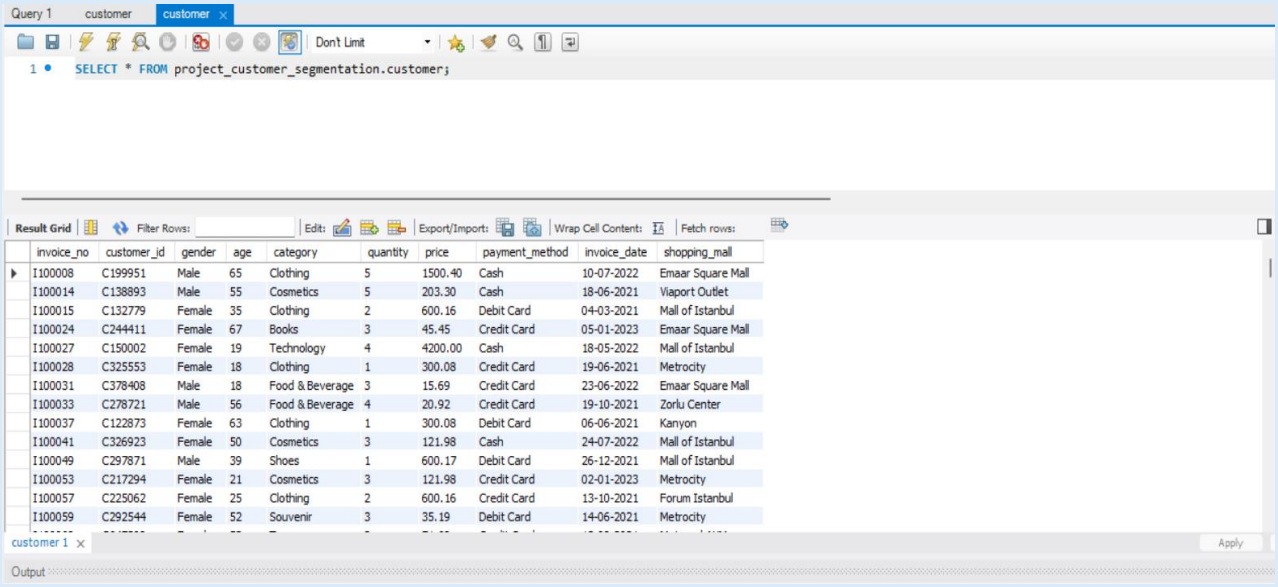
PROJECT OBJECTIVE

- To analyse customer shopping behaviour using Microsoft Power BI.
- To compare shopping patterns between male and female customers.
- To identify which gender generates higher revenue and purchases more products.
- To analyse customer age groups and their shopping preferences.
- To understand product category distribution and payment methods.
- To present key insights through interactive charts and dashboards.

DATASET DESCRIPTION

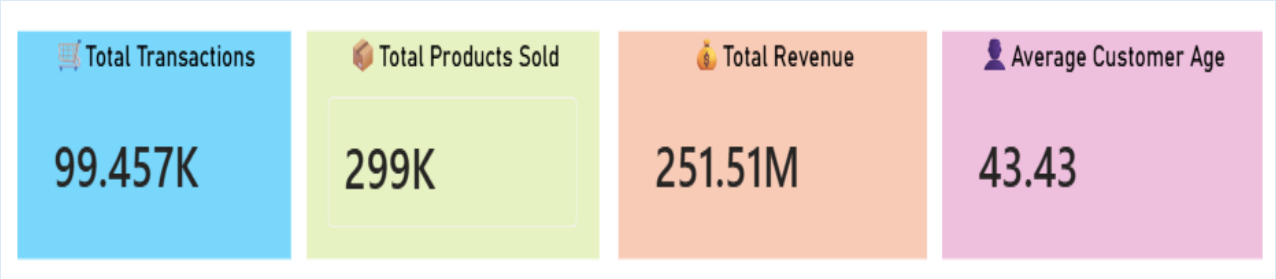
- **Dataset Name:** Customer Shopping Dataset
- **Total Records:** 99,457 shopping transactions
- **Location:** 10 shopping malls in Istanbul, Turkey
- **Time Period:** 2021–2023
- **Customer Details:** Gender and Age
- **Transaction Details:** Invoice Number, Quantity, Price and Payment Method
- **Product Details:** Category and Shopping Mall

Purpose: To analyse customer purchasing patterns, product preferences and revenue using Power BI.



KEY PERFORMANCE INDICATORS (KPIs)

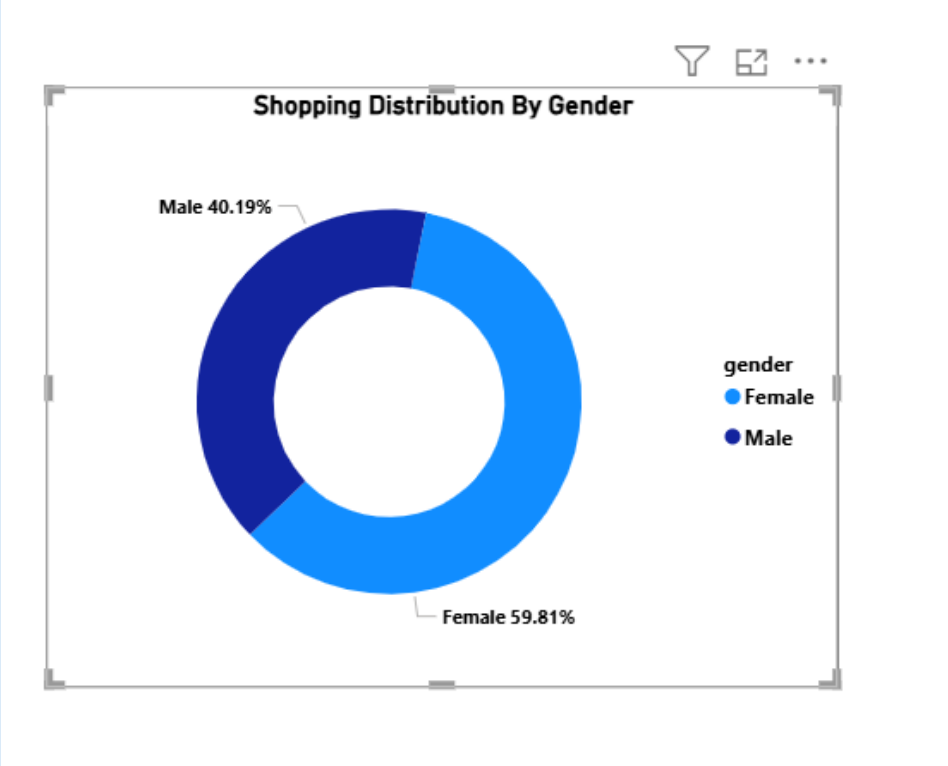
- **Total Customers:** Count of customer transactions
- **Total Revenue:** Sum of sales revenue
- **Total Quantity Sold:** Total number of products purchased
- **Average Purchase Value:** Average spending per transaction
- **Total Product Categories:** Number of distinct product categories
- **Total Shopping Malls:** Number of distinct shopping malls



SHOPPING DISTRIBUTION BY GENDER

- Analysed customer shopping transactions by gender.
- Compared the number of transactions made by male and female customers.
- Identified the gender with the highest shopping activity.
- Used a gender-wise chart to visualise shopping distribution.

Key Insight: Female customers contributed more shopping transactions than male customers in the dataset.



PRODUCTS SOLD AND REVENUE BY GENDER

- Compared the total quantity of products purchased by male and female customers.
- Analysed the revenue generated by each gender.
- Identified which gender purchased more products.
- Compared the revenue contribution of male and female customers.
- Used charts to understand gender-wise sales performance.

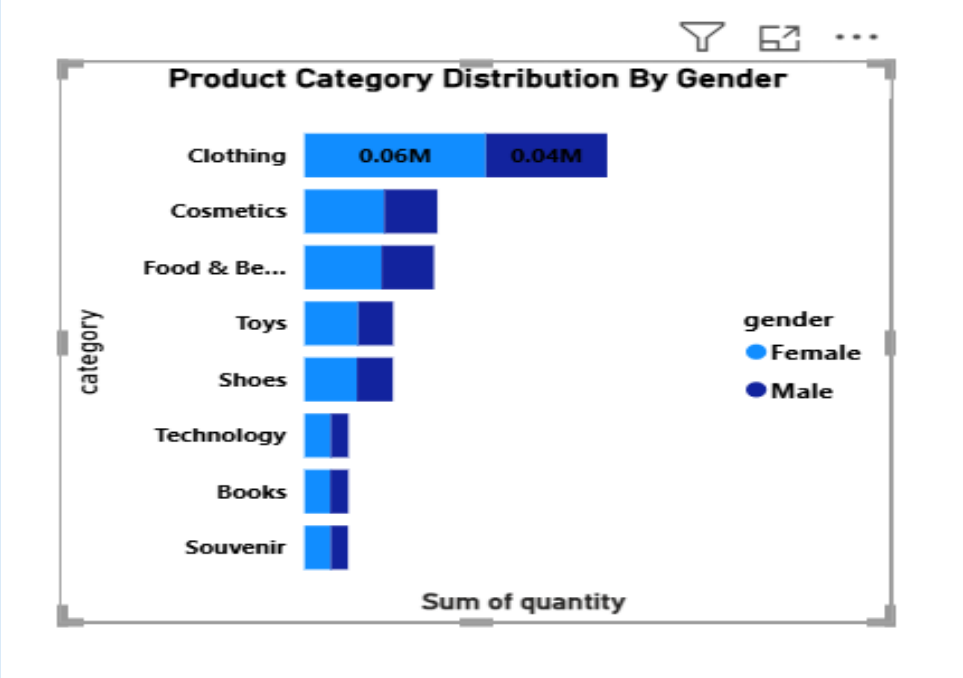
Key Insight: Female customers contributed more purchases and generated higher revenue than male customers in the dataset.



CATEGORY DISTRIBUTION BY GENDER

- Analysed product category preferences among male and female customers.
- Compared gender-wise purchases across different product categories.
- Identified popular categories for each gender.
- Examined differences in shopping preferences between male and female customers.
- Used visualisations to compare category-wise shopping behaviour.

Key Insight: Clothing was the leading purchase category overall. Gender-wise category comparisons help identify differences in customer preferences.



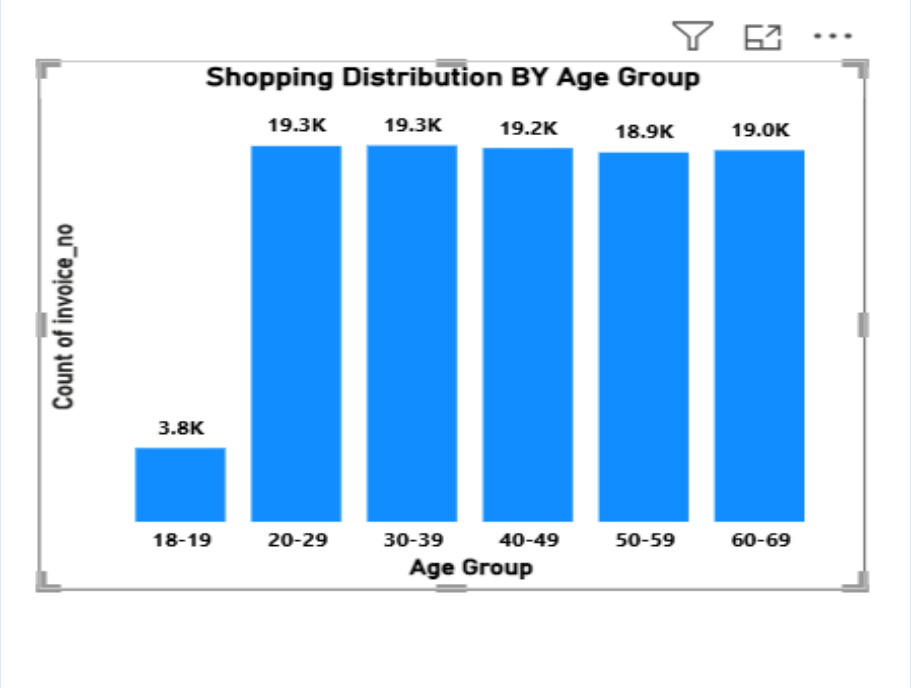
SHOPPING DISTRIBUTION BY AGE GROUP

- Analysed customer shopping transactions across different age groups.
- Divided customers into age groups for comparison.
- Compared shopping activity among different age categories.
- Identified the age group with the highest number of transactions.
- Used visualisations to understand age-wise shopping behaviour.

Age Groups:

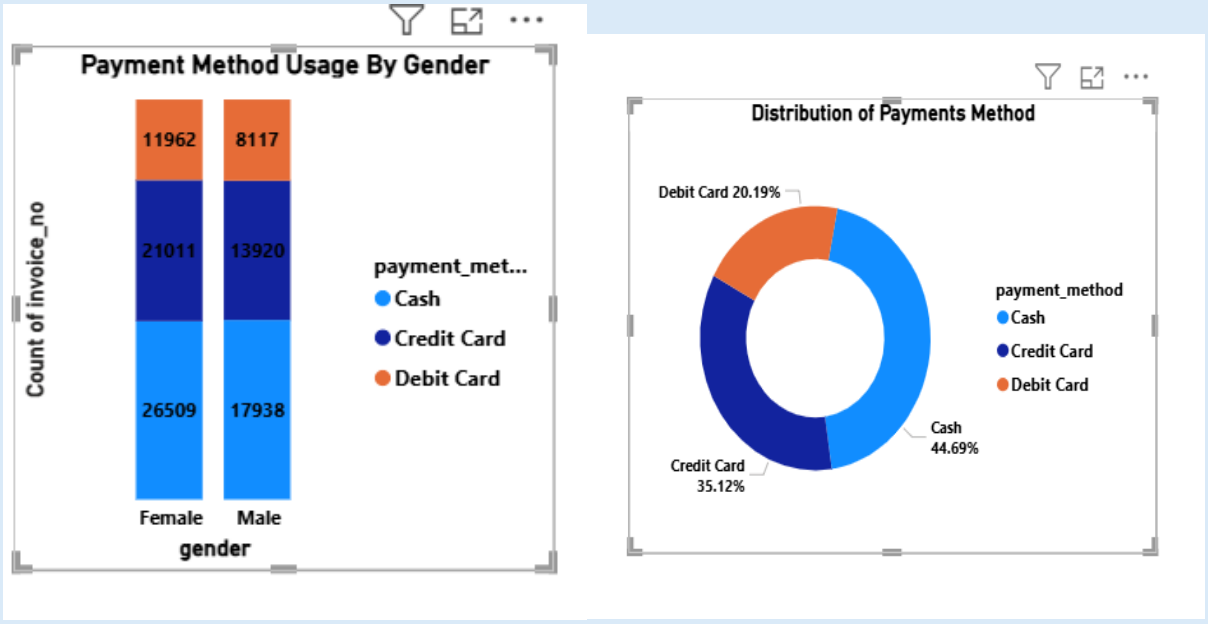
- Under 20
- 20–29
- 30–39
- 40–49
- 50–59
- 60 and above

Key Insight: Age-wise analysis helps identify which customer groups.

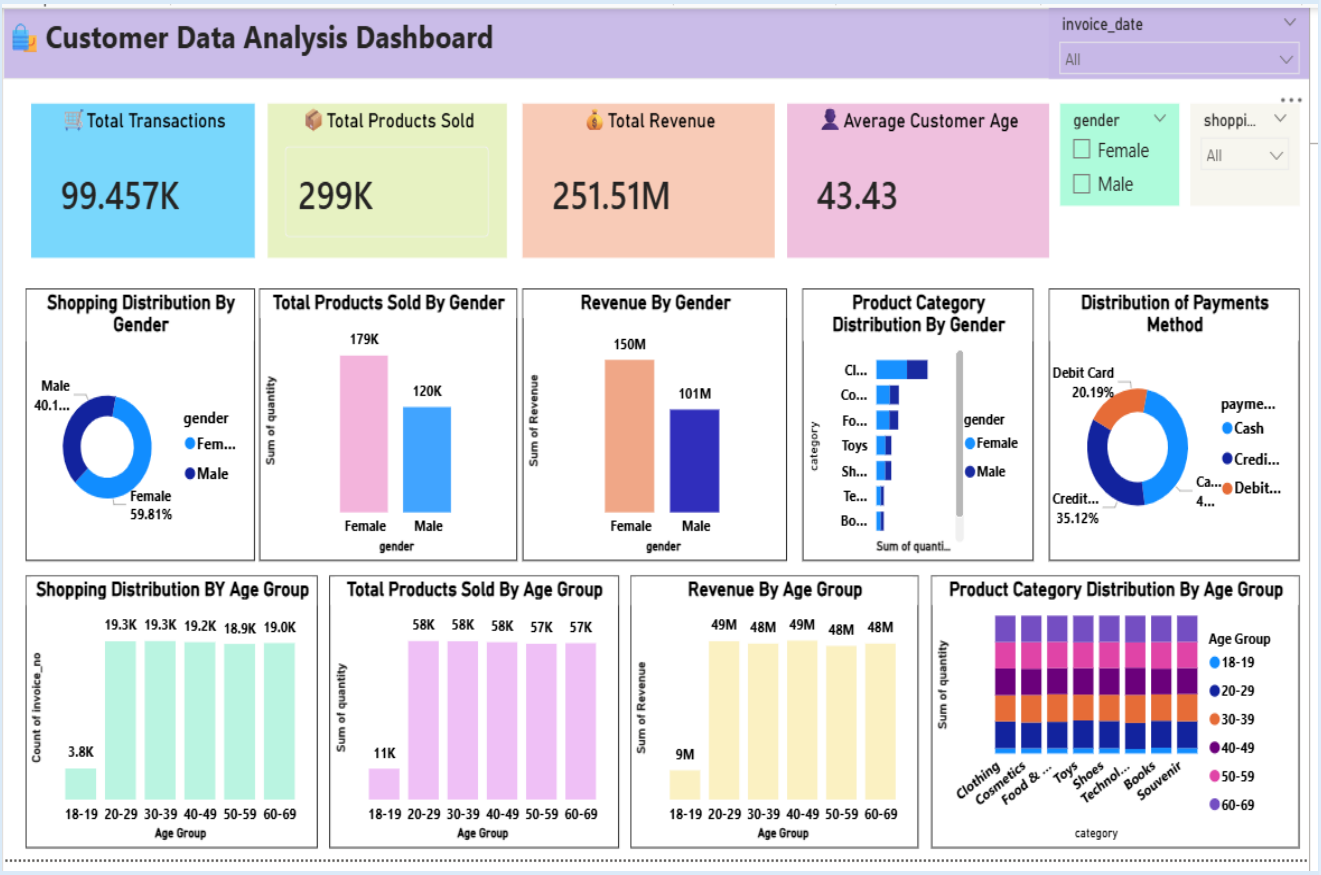


CATEGORY AND PAYMENT METHOD ANALYSIS

- Analysed customer purchases across different product categories.
- Compared the usage of different payment methods.
- Identified the most commonly used payment method.
- Examined customer payment preferences.



CUSTOMER DATA ANALYSIS DASHBOARD



KEY INSIGHTS, RECOMMENDATIONS AND CONCLUSION

Key Insights

- Female customers contributed more shopping transactions and revenue.
- Clothing was the most popular purchase category.
- Cash was the most commonly used payment method.
- Age-group analysis helps identify customer shopping patterns.

Recommendations

- Focus on popular product categories.
- Understand the preferences of different customer groups.
- Use customer insights to improve marketing strategies.

Conclusion

Power BI helps analyse customer shopping behaviour through interactive visualisations and supports data-driven business decisions.